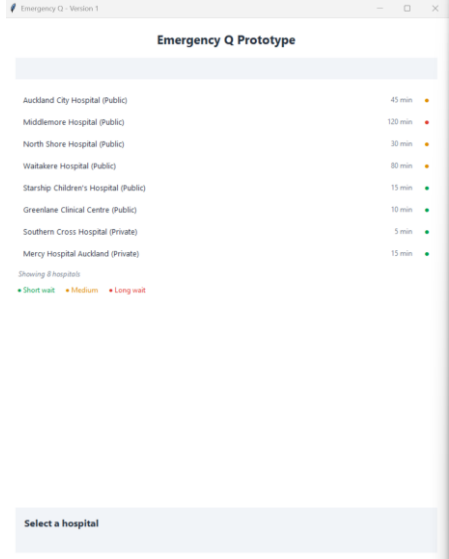


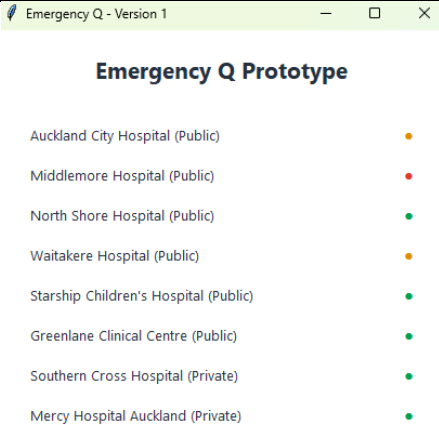
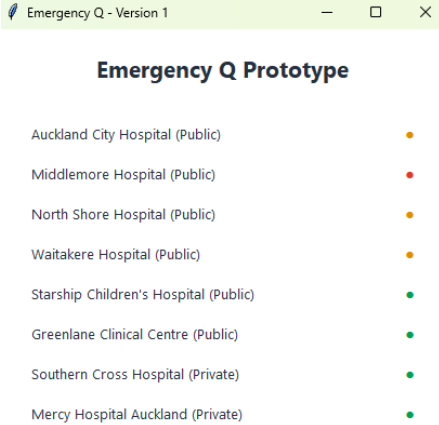
Emergency Q (WaitSmart NZ) - Component Testing Table

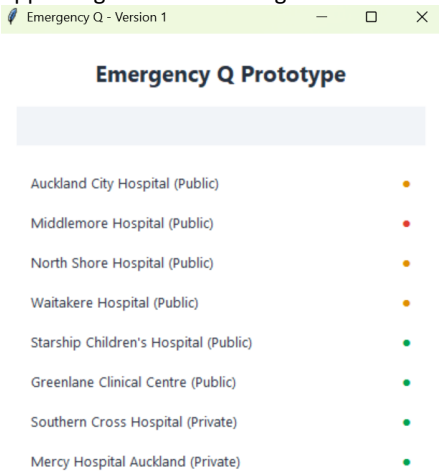
AS91906 / AS91907 - Krish Ram

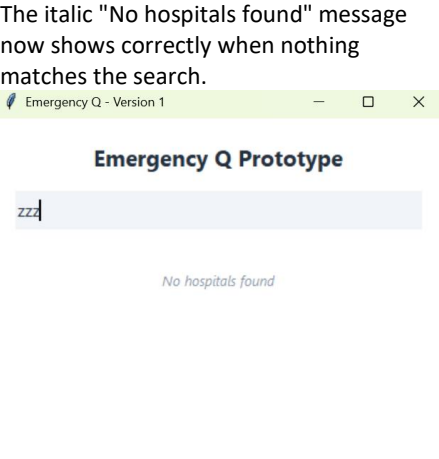
Version 1						
Stage	Input (what did you click, type, or do)	Expected output (what should happen)	Test result (pass/fail)	Test result explanation (did it work? if not, what happened)	Expected, Boundary, or Invalid	Action taken to fix (where needed)
Startup / window	Ran the program after adding the colour constants (Commit 2)	A blank white window titled "Emergency Q - Version 1" opens	Fail	The program crashed with "NameError: name 'root' is not defined". The lines root.geometry(...) and root.configure(bg=WHITE) were placed above root = tk.Tk(), so root was used before it existed. 	Expected	Moved root.geometry() and root.configure(bg=WHITE) to below root = tk.Tk() so root is created before it is used. Re-ran: the white window opened correctly.
Startup / window	Ran the program again after the fix	A blank white window titled "Emergency Q - Version 1" opens with a white background	Pass	The white window opened with the correct title and no errors. 	Expected	None
Hospital list	Added the list frame (list_frame) and ran the program (Commit 5)	The window still opens with the title and no errors. The empty frame is not visible yet.	Pass	The window opened correctly with no errors. Nothing changed visually yet because the frame is empty.	Expected	None

				Auckland City Hospital (Public) Middlemore Hospital (Public) North Shore Hospital (Public) Waitakere Hospital (Public) Starship Children's Hospital (Public) Greenlane Clinical Centre (Public) Southern Cross Hospital (Private) Mercy Hospital Auckland (Private) <i>Showing 8 hospitals</i>		
Hospital list	Added the loop to show each hospital as a row and ran the program (Commit 6)	All 8 hospitals appear in the list, each showing "name (type)".	Fail	The program crashed with "KeyError: 'nmae'" on the line that builds the label text. h["nmae"] asked the hospital dictionary for a key that does not exist. 	Expected	Read the KeyError (it named the missing key 'nmae'), compared it to the real data keys (name, wait, phone, type), and saw it was a typo of "name". Changed h["nmae"] to h["name"].
Hospital list	Re-ran the program after fixing the typo (Commit 6)	All 8 hospitals appear as rows showing "name (type)".	Pass	After changing h["nmae"] to h["name"], all 8 hospitals displayed correctly as a list.	Expected	None (fixed in the previous step)

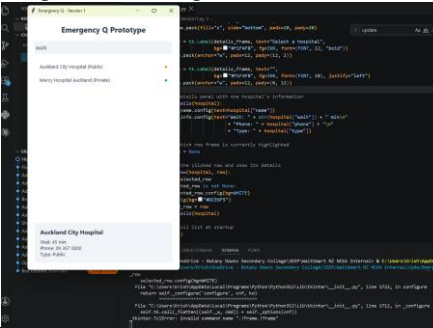
						
Colour coding	Added the wait_colour() function and ran the program (Commit 7)	The program runs with no errors. No visible change yet because nothing uses the function.	Pass	The program ran with no errors. wait_colour is defined but not called yet, so the display is unchanged (same as Commit 6). 	Expected	None
Colour coding (boundary)	Added coloured dots (Commit 8). Checked North Shore Hospital, which has a wait of exactly 30 minutes (the SHORT boundary).	North Shore should show an AMBER dot, because 30 is not under 30 - the green band is only for waits	Fail	North Shore showed a GREEN dot instead of amber. The wait_colour function used "if minutes <= SHORT:", so a wait of exactly 30 was counted as green. This is an off-by-one / boundary error at the SHORT threshold.	Boundary	Changed "if minutes <= SHORT:" to "if minutes < SHORT:" so that exactly 30 minutes falls through to the amber band. Re-ran the program.

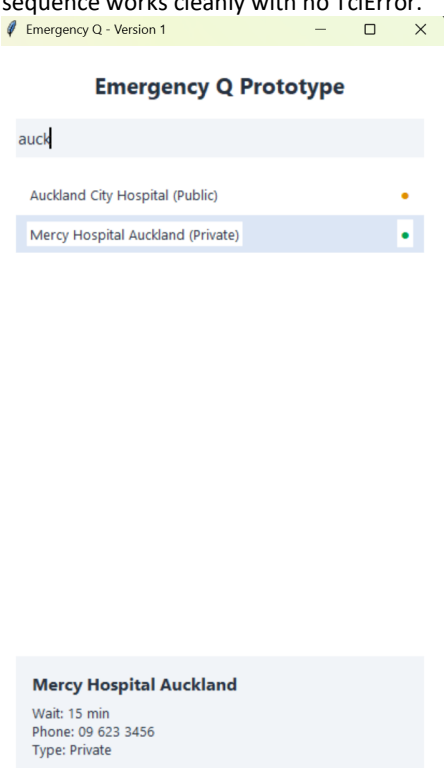
		below 30 minutes.				
Colour coding (boundary)	Re-ran the program after changing <code><=</code> to <code><</code> in <code>wait_colour</code> (Commit 8).	North Shore (30 min) shows AMBER. Waits under 30 stay green, 90+ stay red.	Pass	North Shore now shows amber. All other dots stayed correct: Greenlane (10) and Southern Cross (5) green, Auckland City (45) and Waitakere (80) amber, Middlemore (120) red. The boundary value of exactly 30 is now handled correctly. 	Boundary	None (fixed in the previous step).
Search / filter	Started the program with an empty search box (Commit 9-10).	All 8 hospitals are listed, each with its	Pass	With no search text, <code>update_list()</code> shows all 8 hospitals. Clearing the old rows before rebuilding stops duplicates	Expected	None

		colour dot.		appearing as the text changes.  The screenshot shows a web browser window titled 'Emergency Q - Version 1'. Below the title bar is a header 'Emergency Q Prototype'. Underneath is a search bar. Below the search bar is a list of hospitals, each with a colored dot to its right: - Auckland City Hospital (Public) - Orange dot - Middlemore Hospital (Public) - Red dot - North Shore Hospital (Public) - Orange dot - Waitakere Hospital (Public) - Orange dot - Starship Children's Hospital (Public) - Green dot - Greenlane Clinical Centre (Public) - Green dot - Southern Cross Hospital (Private) - Green dot - Mercy Hospital Auckland (Private) - Green dot		
Search / filter	Typed "auckl" into the search box (Commit 10).	Only hospitals whose name contains "auckl" appear.	Pass	The list filtered live as I typed. Only Auckland City Hospital and Mercy Hospital Auckland were shown, each with the correct colour dot. Matching is case-insensitive because both the query and the name are lowercased.  The screenshot shows the same web browser window. The search bar now contains the text 'auckl'. The list of hospitals is filtered to show only two items: - Auckland City Hospital (Public) - Orange dot - Mercy Hospital Auckland (Private) - Green dot	Expected	None
Search / no results	Typed "zzz" - a value that matches no hospital (Commit 11).	A "No hospitals found" message should appear.	Fail	No message appeared, just a blank space. The check "if matches == 0:" compared the matches LIST to the number 0, which is always False, so the message block never ran.	Invalid	Changed "if matches == 0:" to "if len(matches) == 0:" so it checks whether the list is empty. Re-ran the program.

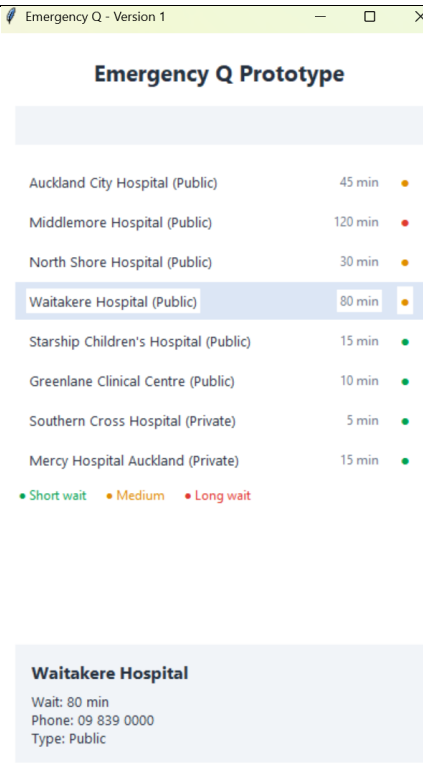
						
Search / no results	Typed "zzz" again after the fix.	The "No hospitals found" message appears.	Pass	The italic "No hospitals found" message now shows correctly when nothing matches the search. 	Invalid	None (fixed in the previous step).
Details panel / click	Clicked several different hospitals in turn (Commit 14).	The details panel should show the wait, phone and type of the	Fail	Every hospital I clicked showed "Mercy Hospital Auckland" (the last one in the list). The click handler was "lambda event: show_details(h)" - the lambda captured the loop variable h by reference, so after	Expected	Captured the value at creation time with a default argument: "lambda event, h=h: show_details(h)". Each row now stores its own hospital.

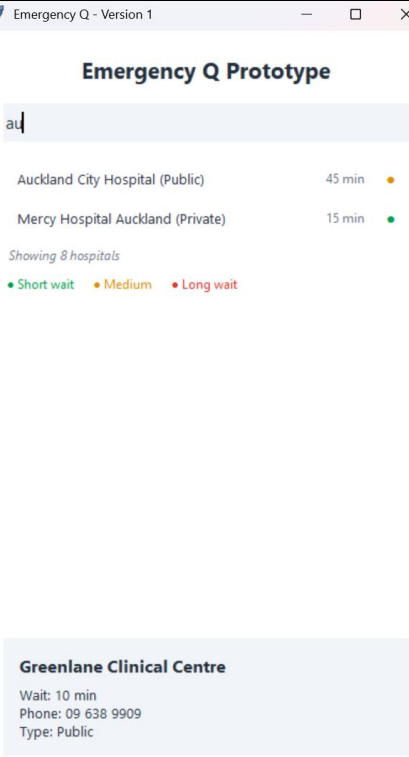
		hospital I clicked.		the loop finished all rows pointed at the final h (Mercy). Emergency Q - Version 1 Emergency Q Prototype Auckland City Hospital (Public) Middlemore Hospital (Public) North Shore Hospital (Public) Waitakere Hospital (Public) Starship Children's Hospital (Public) Greenlane Clinical Centre (Public) Southern Cross Hospital (Private) Mercy Hospital Auckland (Private) Mercy Hospital Auckland Wait: 15 min Phone: 09 623 3456 Type: Private		
Details panel / click	Clicked Starship Children's Hospital after the fix (Commit 15).	The panel shows Starship Children's Hospital with Wait 15 min, Phone 09 367 0000, Type Public.	Pass	After the h=h fix, clicking a hospital shows that hospital's own details. Clicking Starship correctly showed Wait: 15 min, Phone: 09 367 0000, Type: Public.	Expected	None (fixed in the previous step).

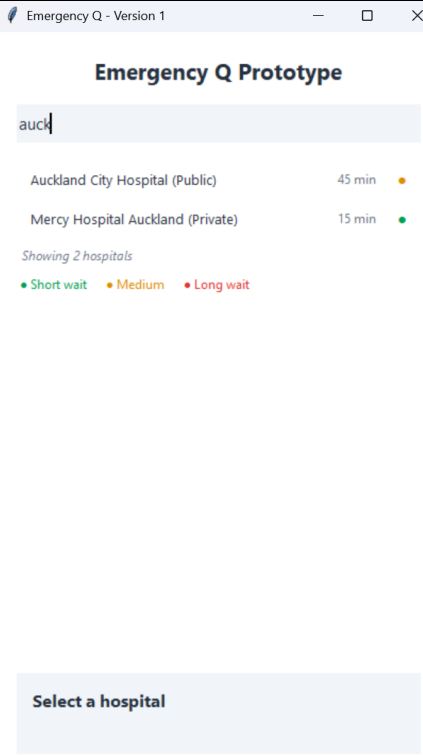
						
Selection + search	Clicked a hospital to highlight it, then typed in the search box, then clicked a hospital in the filtered list (Commit 16).	The new hospital highlights and its details show, with no errors.	Fail	The program crashed with "<code>_tkinter.TclError: invalid command name .!frame.!frame</code>". Typing in the search box made <code>update_list()</code> destroy the old rows, but <code>selected_row</code> still pointed at the destroyed row. The next click called <code>selected_row.config(bg=WHITE)</code> on a widget that no longer existed. 	Invalid	Added " <code>global selected_row</code> " and " <code>selected_row = None</code> " at the top of <code>update_list()</code> so the remembered selection is cleared whenever the list rebuilds. <code>select_row</code> then skips the un-highlight step when nothing is selected.
Selection +	Repeated the click -> search -	The new	Pass	After resetting <code>selected_row</code> when the list	Invalid	None (fixed in the previous step).

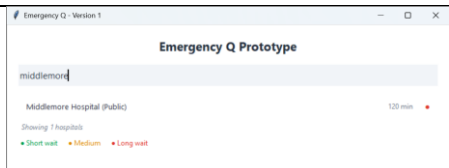
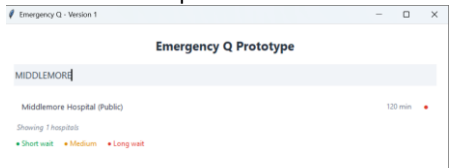
search	> click sequence after the fix.	hospital highlights and its details show, with no errors in the terminal.		refreshes, the click -> search -> click sequence works cleanly with no TclError. 		
Wait time on rows	Ran the program after adding the wait-time label to each row (Commit 17).	Each hospital row shows its wait time as text, e.g. "45 min", next to the colour dot.	Fail	The program crashed with "TypeError: unsupported operand type(s) for +: 'int' and 'str'" on the line <code>text=h["wait"] + " min"</code>. <code>h["wait"]</code> is a number and " min" is text, and Python will not join a number to a string with +. 	Invalid	Wrapped the number in <code>str()</code> : <code>text=str(h["wait"]) + " min"</code> . (Same fix I had used earlier in <code>show_details</code> on Day 4.)
Wait time on rows	Re-ran the program after wrapping the wait in <code>str()</code> (Commit 17).	Each row shows its wait time as text next to the dot.	Pass	Every hospital row now shows its wait time, e.g. "45 min", to the left of the colour dot, with no errors.	Expected	None (fixed in the previous step).

				Emergency Q - Version 1 Emergency Q Prototype Auckland City Hospital (Public) 45 min Middlemore Hospital (Public) 120 min North Shore Hospital (Public) 30 min Waitakere Hospital (Public) 80 min Starship Children's Hospital (Public) 15 min Greenlane Clinical Centre (Public) 10 min Southern Cross Hospital (Private) 5 min Mercy Hospital Auckland (Private) 15 min Auckland City Hospital Wait: 45 min Phone: 09 367 0000 Type: Public		
Colour legend	Ran the program after adding the colour legend (Commit 18).	A small key appears showing a green "Short wait", amber "Medium" and red "Long wait" so users know what the dots mean.	Pass	The legend displayed correctly under the search box with the three coloured items, matching the dot colours used in the list.	Expected	None

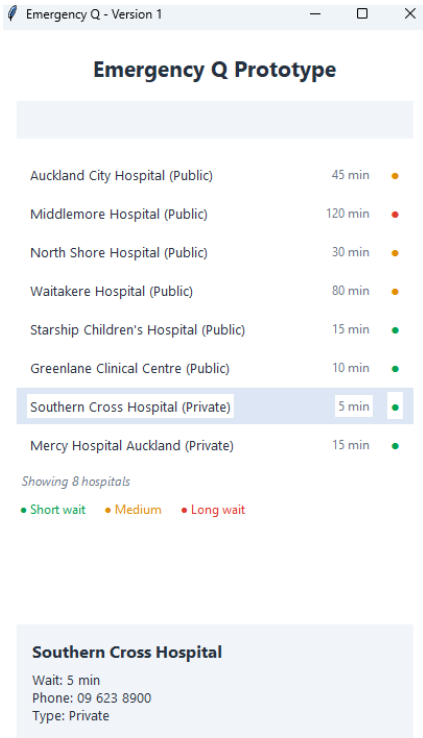
						
Result count	Searched "au" after adding the count label (Commit 19). The list filtered to 2 hospitals.	The count should read "Showing 2 hospitals".	Fail	The count stayed on "Showing 8 hospitals" no matter what I searched. The line used len(HOSPITALS) - the master list, which is always 8 - instead of len(matches), the filtered list.	Boundary	Changed len(HOSPITALS) to len(matches) so the count reflects the filtered results.

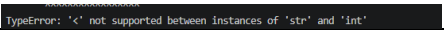
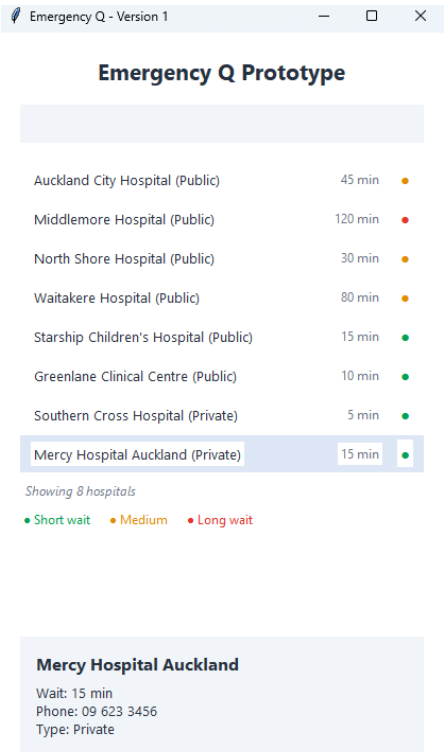
						
Result count	After the fix, searched "auck" (2 matches), cleared the box (8), and searched "zzz" (0 matches).	"Showing 2 hospitals", then "Showing 8 hospitals", then "Showing 0 hospitals".	Pass	The count now updates with the search: "auck" gave Showing 2 hospitals, an empty box gave Showing 8, and "zzz" gave Showing 0 with the "No hospitals found" message. Boundary values of 2, 8 and 0 all correct.	Boundary	None (fixed in the previous step).

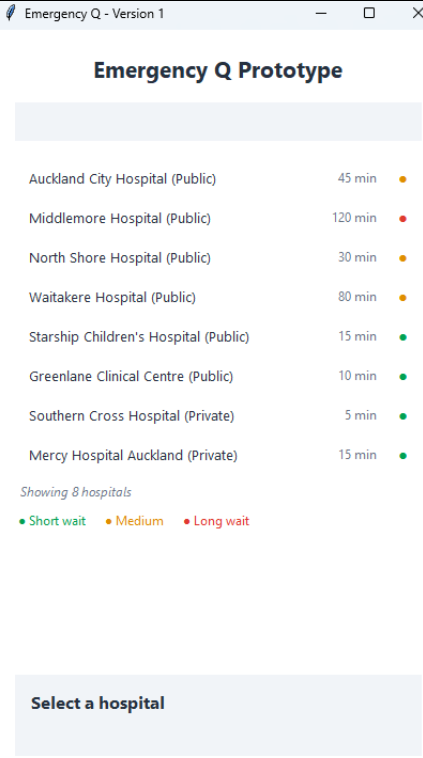
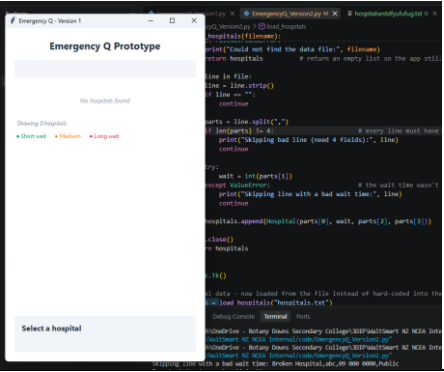
				 The screenshot shows a web browser window titled 'Emergency Q - Version 1'. The page header is 'Emergency Q Prototype'. A search bar contains the text 'auck'. Below the search bar, two results are listed: 'Auckland City Hospital (Public)' with a wait time of 45 min and a red dot, and 'Mercy Hospital Auckland (Private)' with a wait time of 15 min and a green dot. Below the list, it says 'Showing 2 hospitals'. At the bottom, there is a legend: 'Short wait' (green dot), 'Medium' (yellow dot), and 'Long wait' (red dot). A button at the bottom says 'Select a hospital'.		
Search (many matches)	Searched "hospital" (final regression test).	Every hospital whose name contains "Hospital" appears - 7 of them - and the count reads "Showing 7 hospitals".	Pass	Searching "hospital" returned the 7 matching hospitals with the count at "Showing 7 hospitals". Greenlane Clinical Centre was correctly left out because its name has no "Hospital" in it.  The screenshot shows the same web browser window. The search bar contains the text 'hospital'. Below the search bar, seven results are listed: 'Auckland City Hospital (Public)' (45 min, red dot), 'Middlemore Hospital (Public)' (120 min, red dot), 'North Shore Hospital (Public)' (30 min, yellow dot), 'Waiatare Hospital (Public)' (80 min, yellow dot), 'Starship Children's Hospital (Public)' (15 min, green dot), 'Southern Cross Hospital (Private)' (5 min, green dot), and 'Mercy Hospital Auckland (Private)' (15 min, green dot). Below the list, it says 'Showing 7 hospitals'. The legend at the bottom is the same.	Expected	None
Search (single match)	Searched "middlemore" (final regression test).	Only Middlemore Hospital shows and the count reads "Showing 1	Pass	Exactly one result with the count at 1 - the lower boundary of a non-empty result list.	Boundary	None

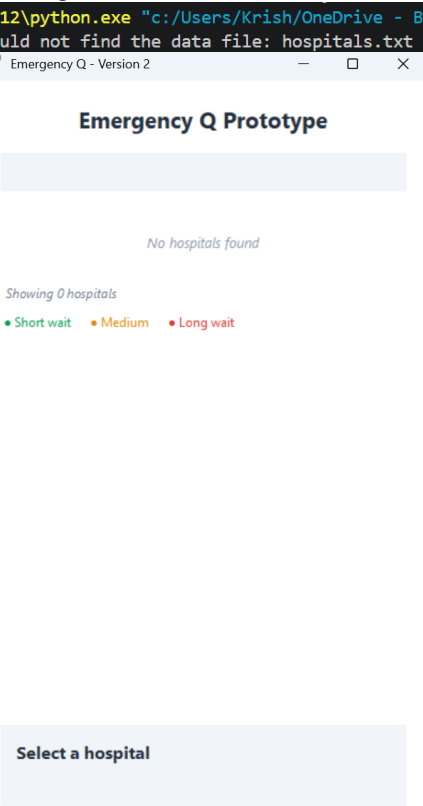
		hospitals".				
Search (case-insensitive)	Searched "MIDDLEMORE" in capitals (final regression test).	Middlemore is still found even though it is typed in capitals.	Pass	The query and the hospital name are both lowercased before comparing, so capitals still match. "MIDDLEMORE" returned Middlemore Hospital. 	Invalid	None
Search (substring)	Searched "shore" - a word from the middle of a name (final regression test).	North Shore Hospital appears; matching is not limited to the start of the name.	Pass	"in" checks for a substring anywhere in the name, so "shore" matched "North Shore Hospital". 	Invalid	None
Colour coding (all bands)	Checked every hospital dot against its wait band (final regression test).	Green under 30, amber 30-89, red 90 and over.	Pass	All 8 dots match their wait bands: green = Starship 15, Greenlane 10, Southern Cross 5, Mercy 15; amber = Auckland 45, North Shore 30, Waitakere 80; red = Middlemore 120. This includes the exact 30-minute boundary (North Shore = amber) and the 90-minute cut-off (a wait of exactly 90 returns red).	Boundary	None

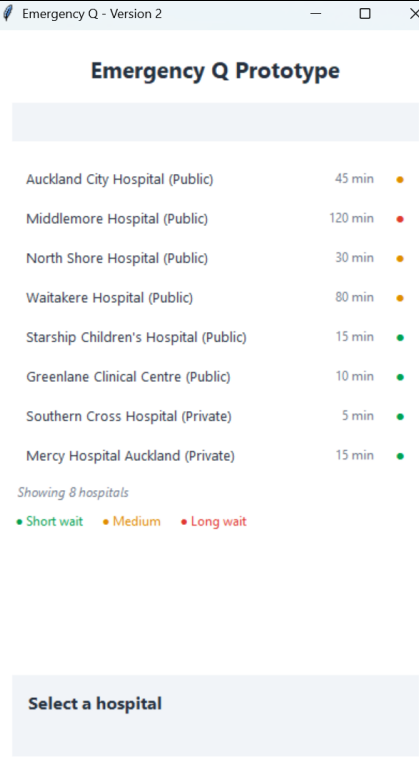
				Emergency Q - Version 1 Emergency Q Prototype Auckland City Hospital (Public) 45 min Middlemore Hospital (Public) 120 min North Shore Hospital (Public) 30 min Waitakere Hospital (Public) 80 min Starship Children's Hospital (Public) 15 min Greenlane Clinical Centre (Public) 10 min Southern Cross Hospital (Private) 5 min Mercy Hospital Auckland (Private) 15 min Auckland City Hospital Wait: 45 min Phone: 09 367 0000 Type: Public		
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Version 2						
Stage	Input (what did you click, type, or do)	Expected output (what should happen)	Test result (pass/fail)	Test result explanation (did it work? if not, what happened)	Expected, Boundary, or Invalid	Action taken to fix (where needed)
Data model (dictionary -> class)	Ran Version 2 after converting the hospital data to a Hospital class (Commit 1).	The list of 8 hospitals appears, now powered by Hospital objects instead of dictionaries.	Fail	Crashed with "TypeError: 'Hospital' object is not subscriptable". The data had been changed to Hospital objects but the code still used dictionary syntax like h["name"]. Objects are not accessed with ["..."]. 	Expected	Changed every dictionary access to attribute access: h["name"] -> h.name, h["type"] -> h.kind, h["wait"] -> h.wait, and the same inside show_details.
Data model (dictionary -> class)	Re-ran Version 2 after converting all accesses to attribute style (Commit 1).	All 8 hospitals list correctly and clicking one shows its details.	Pass	The app runs the same as V1 but is now powered by the Hospital class. Clicking Southern Cross Hospital correctly showed Wait: 5 min, Phone: 09 623 8900, Type: Private. 	Expected	None (fixed in the previous step).
File loading	Ran Version 2 after switching HOSPITALS to	The 8 hospitals load from the	Fail	Two problems. First a FileNotFoundError because the terminal was not in the code	Invalid	Ran the program from inside the code folder, and wrapped the wait in int(): wait

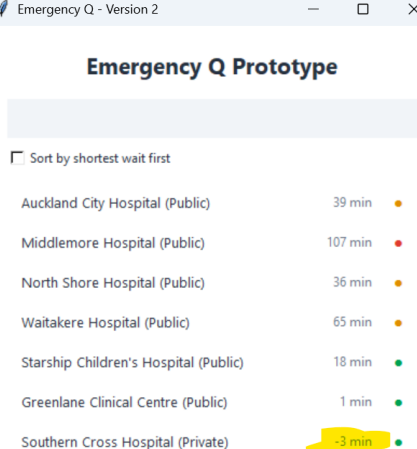
	load_hospitals("hospitals.txt") (Commit 2).	file with correct wait times and colours.		folder. Then, once found, it crashed with "TypeError: '<' not supported between instances of 'str' and 'int'" - line.split(",") returns strings, so parts[1] was the text "45", and status_colour compared a string to a number. 		= int(parts[1]).
File loading	Re-ran from the code folder with wait = int(parts[1]) (Commit 2).	All 8 hospitals load from hospitals.txt and clicking one shows its details.	Pass	The app loaded all 8 hospitals from the file with correct wait times and colours. Clicking Mercy Hospital Auckland showed Wait: 15 min, Phone: 09 623 3456, Type: Private. 	Expected	None (fixed in the previous step).
Robustness : bad line	Added a line with a non-number wait (Broken Hospital,abc,09 000 0000,Public) to hospitals.txt and ran the app (Commit 3).	The bad line is skipped and the 8 real hospitals still show, with no crash.	Pass	The app opened with the normal 8 hospitals and the terminal printed "Skipping line with a bad wait time: Broken Hospital,abc,...". The try/except around int() caught the bad value instead of crashing.	Invalid	None - handled by the robust loader.

						
Robustness : missing file	Renamed hospitals.txt so it could not be found, then ran the app (Commit 3).	The app opens with an empty list instead of crashing.	Pass	The window opened showing "No hospitals found" and "Showing 0 hospitals", and the terminal printed "Could not find the data file". The try/except around open() returned an empty list so the program still ran. 	Invalid	None - handled by the robust loader. (Renamed the file back afterwards.)
App class: NameError	Ran Version 2 after wrapping the whole program in the	The app opens and loads the 8	Fail	NameError: name 'hospitals' is not defined. Inside update_list the code said	Expected	Changed it to self.hospitals.

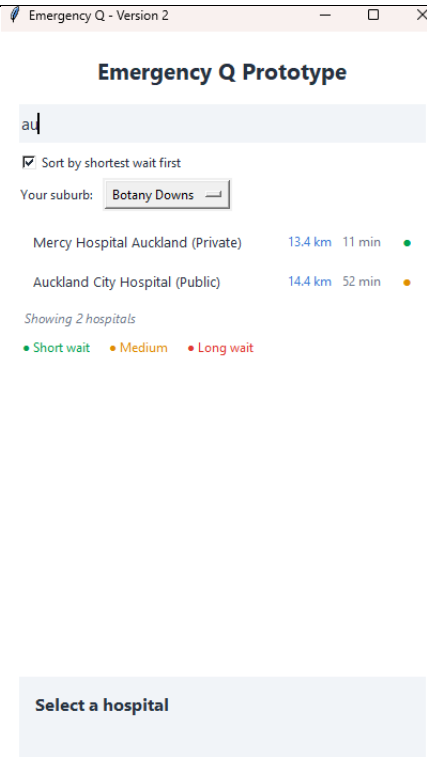
	App class (Commit 4).	hospitals, now from inside the App class.		"hospitals", but the data is stored on the object as self.hospitals, so it needed the self. prefix. <pre> matches = [h for h in hospitals if query in h.name.lower()]] AAAAAAAA NameError: name 'hospitals' is not defined. Did you mean: 'self.hospitals'? </pre>		
App class: data not loading	Re-ran after fixing self.hospitals, using the Run button from the project root (Commit 4).	All 8 hospitals load from the file.	Fail	The app opened but showed "Showing 0 hospitals". The terminal printed "Could not find the data file: hospitals.txt" - it was run from the project root, not the code folder, so the relative path missed the file. The robust loader caught it and returned an empty list instead of crashing. <pre> h312\python.exe "c:/Users/Krish/OneDrive - B Could not find the data file: hospitals.txt </pre> 	Invalid	Loaded the file by an absolute path built from the script location: BASE_DIR = os.path.dirname(os.path.abspath(__file__)), then load_hospitals(os.path.join(BASE_DIR, "hospitals.txt")). Now it finds the file from any folder.
App class: working	Re-ran with the absolute-path load (Commit 4).	All 8 hospitals load with correct wait times and colours, from any run location.	Pass	The full app now runs inside the App class and loads all 8 hospitals correctly regardless of which folder it is run from. Search, click, highlight and count all still work.	Expected	None.

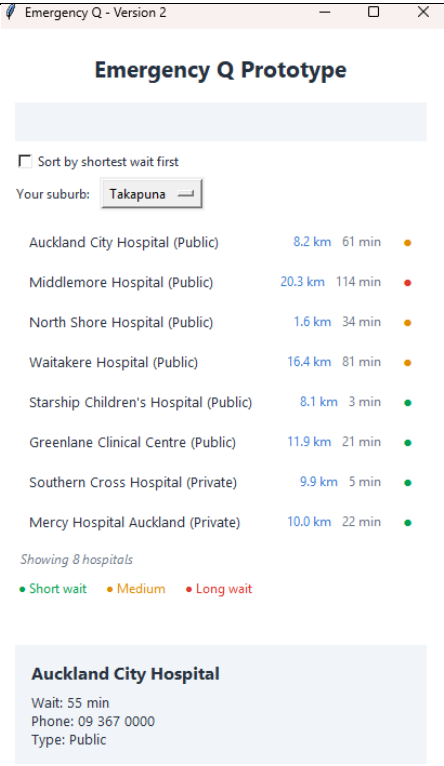
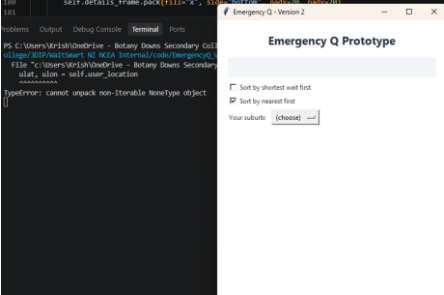
						
Sort by wait time	Ticked "Sort by shortest wait first" (Commit 5).	The shortest wait should be at the top (Southern Cross, 5 min) and the longest at the bottom (Middlemore, 120 min).	Fail	The list sorted the opposite way - longest first: Middlemore (120) at the top and Southern Cross (5) at the bottom. matches.sort was called with reverse=True, which sorts descending.	Expected	Removed reverse=True so it sorts ascending (shortest first): matches.sort(key=lambda h: h.wait).

						
Sort by wait time	Ticked the sort box again after removing reverse=True (Commit 5).	Shortest wait at the top, longest at the bottom.	Pass	The list now sorts shortest-first: Southern Cross 5, Greenlane 10, Starship 15, Mercy 15, North Shore 30, Auckland 45, Waitakere 80, Middlemore 120. Unticking the box returns the list to file order.	Expected	None (fixed in the previous step).

						
Live wait times	Ran V2 with the live random wait times and watched for ~15 seconds (Commit 6).	Each wait drifts up and down over time but stays realistic (0 minutes or more).	Fail	After a few seconds Southern Cross showed "-3 min" - a negative wait time. $h.wait = h.wait + random.randint(-6, 6)$ had no lower limit, so the random drift pushed a small wait below zero. 	Boundary	Clamped the value with $\max(0, \dots)$: $h.wait = \max(0, h.wait + random.randint(-6, 6))$, so it can never go below 0.

Live wait times	Re-ran after clamping with <code>max(0, ...)</code> and watched the low-wait hospitals (Commit 6).	Waits change over time but never drop below 0 minutes.	Pass	The live times now drift up and down and bottom out at 0 min (green) instead of going negative. The dots re-colour as waits cross the 30 and 90 minute thresholds.  The screenshot shows a web application titled "Emergency Q Prototype". It has a search bar at the top. Below the search bar, there is a checkbox labeled "Sort by shortest wait first" which is checked. A list of hospitals is displayed with their names, wait times, and status dots. The status dots are green for wait times of 0, 2, and 8 minutes, yellow for 22, 34, and 71 minutes, and red for 108 and 122 minutes. A legend at the bottom indicates that green dots represent "Short wait", yellow dots represent "Medium", and red dots represent "Long wait". A button labeled "Select a hospital" is at the bottom of the list.	Boundary	None (fixed in the previous step).
Location / distance	Picked a suburb from the dropdown - the km distances only appeared after I also typed in the search box (Commit 8).	Picking a suburb should show the distance on each hospital row straight away.	Fail	<code>on_location</code> stored the location but did not refresh the list, so distances only appeared when another action (typing) triggered <code>update_list</code>. The bug was masked whenever a search was active.	Expected	Added <code>self.update_list(self.search_var.get())</code> to <code>on_location</code> so the list redraws as soon as a suburb is picked.

						
Location / distance	Picked Takapuna with an empty search box after the fix (Commit 8).	Distances appear immediately and are realistic for Auckland.	Pass	Distances now show as soon as a suburb is chosen. Takapuna showed North Shore Hospital nearest at 1.6 km and Middlemore farthest at 20.3 km, which matches real geography.	Expected	None (fixed in the previous step).

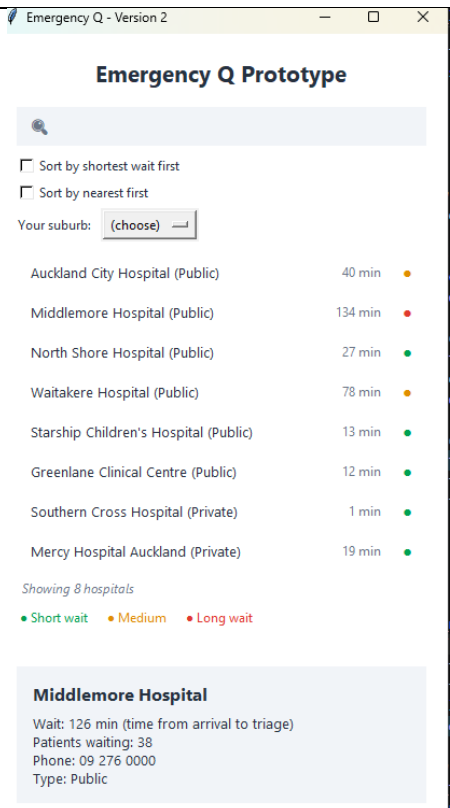
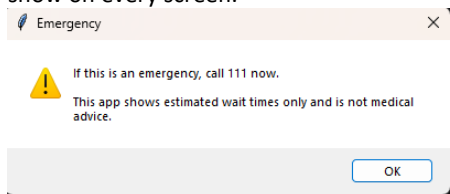
						
Sort by nearest	Ticked "Sort by nearest first" BEFORE choosing a suburb (Commit 9, test 2).	It should sort by distance, or safely do nothing if no suburb is chosen.	Fail	Crashed with "TypeError: cannot unpack non-iterable NoneType object" at ulat, ulon = self.user_location. No suburb was chosen, so self.user_location was still None and could not be unpacked. 	Invalid	Guarded the sort: elif self.nearest_var.get() and self.user_location is not None. Now it does nothing (no crash) until a suburb is picked.
Sort by nearest	Picked Botany Downs, then ticked "Sort by nearest first" (Commit 9, test 1).	Hospitals ordered closest-first.	Pass	The list sorted by distance ascending: Middlemore 7.4 km first, through to Waitakere 25.4 km last.	Expected	None.

				Emergency Q - Version 2 Emergency Q Prototype ☐ Sort by shortest wait first ☒ Sort by nearest first Your suburb: Botany Downs Middlemore Hospital (Public)7.4 km118 min Greenlane Clinical Centre (Public)11.1 km23 min Southern Cross Hospital (Private)13.0 km12 min Mercy Hospital Auckland (Private)13.4 km2 min Auckland City Hospital (Public)14.4 km40 min Starship Children's Hospital (Public)14.5 km18 min North Shore Hospital (Public)20.8 km17 min Waitakere Hospital (Public)25.4 km78 min Showing 8 hospitals Short wait Medium Long wait Select a hospital		
Sort by nearest	Ticked "Sort by nearest first" with no suburb chosen, after adding the guard (Commit 9, test 2 re-run).	No crash; the list stays in its normal order until a suburb is chosen.	Pass	With the guard added, ticking nearest without a suburb leaves the list unchanged and does not crash.	Invalid	None (fixed above).

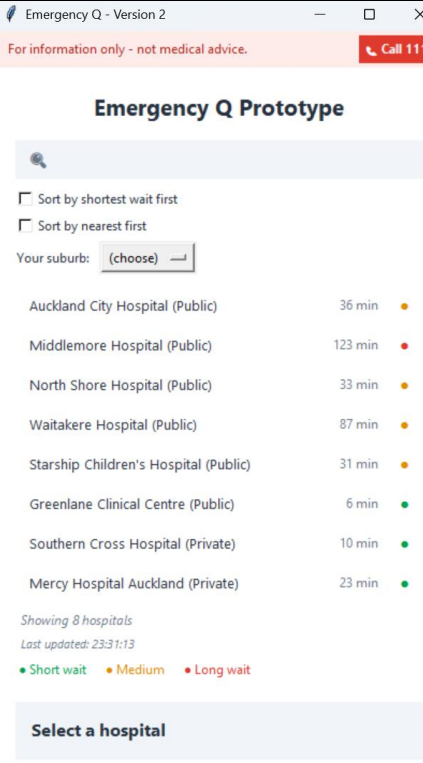
						
Sort by nearest	Ticked BOTH sort boxes at once (Commit 9).	One clear sort order.	Pass	The list sorted by wait time - the wait sort takes priority over nearest because of the if/elif order, so nearest is ignored when both are ticked. Not a crash, but a UX ambiguity.	Invalid	Noted as a known limitation to fix in Version 3: ticking one toggle will untick the other (mutually exclusive).

				Emergency Q - Version 2 Emergency Q Prototype ☒ Sort by shortest wait first ☒ Sort by nearest first Your suburb: Botany Downs Mercy Hospital Auckland (Private)13.4 km2 min Southern Cross Hospital (Private)13.0 km12 min North Shore Hospital (Public)20.8 km17 min Starship Children's Hospital (Public)14.5 km18 min Greenlane Clinical Centre (Public)11.1 km23 min Auckland City Hospital (Public)14.4 km40 min Waitakere Hospital (Public)25.4 km78 min Middlemore Hospital (Public)7.4 km118 min Showing 8 hospitals Short wait Medium Long wait Select a hospital		
Search box icon	Added a magnifier icon inside the search box and ran the program (Commit 10).	A search icon should appear inside the box so it is obvious it is a search field.	Pass	The magnifier icon shows on the left of the search box and typing still filters the list correctly.	Expected	None.

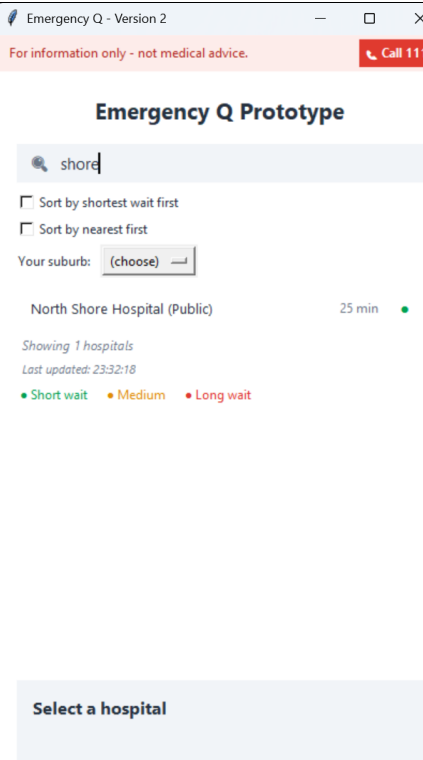
						
Patient count / data fields	Added a patients column to hospitals.txt (now 7 fields) and ran the program (Commit 11).	All 8 hospitals load and each shows its patient count in the details panel.	Fail	The list showed "Showing 0 hospitals / No hospitals found" and the terminal printed "Skipping bad line (need 6 fields)" for every line. hospitals.txt had 7 fields but load_hospitals still checked len(parts) != 6 and Hospital.__init__ only took 6 arguments, so every line was rejected.	Invalid	Updated the loader to require 7 fields and parse patients = int(parts[6]); added patients to Hospital.__init__ and passed it in.
Patient count / data fields	Re-ran after updating the loader and Hospital class to 7 fields, then clicked a hospital (Commit 11).	All 8 hospitals load and the details panel shows the patient count and the wait definition.	Pass	All 8 hospitals loaded again. The details panel now shows "Wait: X min (time from arrival to triage)", "Patients waiting: N", phone and type - matching the inquiry requirement.	Expected	None (fixed above).

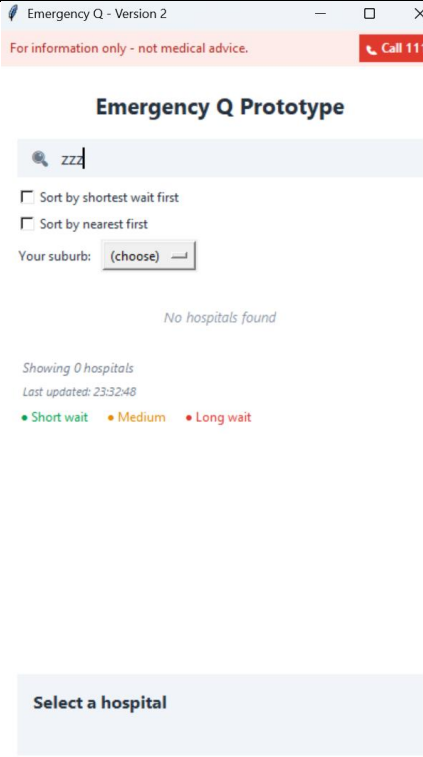
						
Safety banner / 111 button	Added the safety banner and "Call 111" button, then clicked the button (Commit 12).	A warning dialog appears reminding the user to call 111.	Fail	Nothing popped up. The terminal printed "NameError: name 'messagebox' is not defined" from call_111. messagebox is a separate tkinter submodule, so "import tkinter as tk" does not import it. 	Expected	Added "from tkinter import messagebox" to the imports.
Safety banner / 111 button	Re-ran after adding "from tkinter import messagebox" and clicked "Call 111" again (Commit 12).	The warning dialog appears.	Pass	The "Emergency" dialog now appears with "If this is an emergency, call 111 now" and the not-medical-advice reminder. The banner and 111 button show on every screen. 	Expected	None (fixed above).

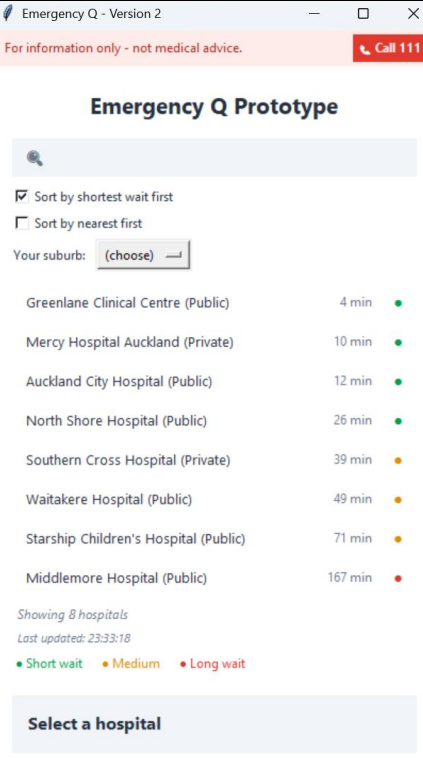
Last updated timestamp	Added a "Last updated" time under the list and watched it while the live wait times refreshed (Commit 13).	A timestamp shows when the data was last refreshed and updates every few seconds.	Pass	The label shows "Last updated: HH:MM:SS" (e.g. 15:24:20) and ticks over each time the live wait times refresh, so the user can see the data is current. Meets the inquiry requirement for a data timestamp. 	Expected	None.
Launch & layout	Full V2 testing pass - launched the finished program and checked the whole screen.	Banner, title, search box, both sort toggles, suburb picker, 8 hospitals with coloured dots, count, last-updated time, legend and an	Pass	Everything rendered correctly: red safety banner + Call 111 at the top, all 8 hospitals with dots, "Showing 8 hospitals", "Last updated: 23:31:13", the legend, and "Select a hospital" in the details panel.	Expected	None.

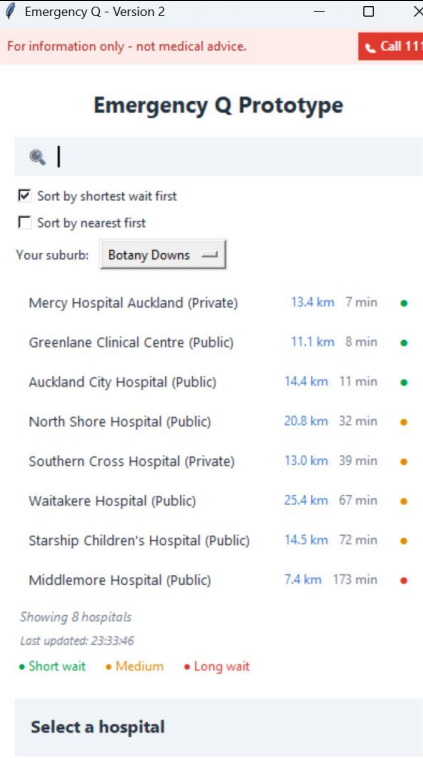
		empty details panel all appear.				
Live wait times	Watched the list for about 40 seconds without touching anything.	Wait times drift up and down every few seconds, dots recolour, and the "Last updated" time changes.	Pass	The waits changed on their own (e.g. Auckland 36 -> 27 min, Middlemore 123 -> 157 min) and "Last updated" moved from 23:31:13 to 23:31:50. No value went negative.	Expected	None.

				Emergency Q - Version 2 For information only - not medical advice. Call 111 Emergency Q Prototype ☐ Sort by shortest wait first ☐ Sort by nearest first Your suburb: (choose) Auckland City Hospital (Public) 27 min Middlemore Hospital (Public) 157 min North Shore Hospital (Public) 25 min Waitakere Hospital (Public) 78 min Starship Children's Hospital (Public) 40 min Greenlane Clinical Centre (Public) 3 min Southern Cross Hospital (Private) 18 min Mercy Hospital Auckland (Private) 13 min Showing 8 hospitals Last updated: 23:31:50 Short wait Medium Long wait Select a hospital		
Search - partial match	Typed "shore" into the search box.	Only hospitals whose name contains "shore" show (case-insensitive substring match).	Pass	The list filtered to just North Shore Hospital and the count read "Showing 1 hospitals".	Expected	None.

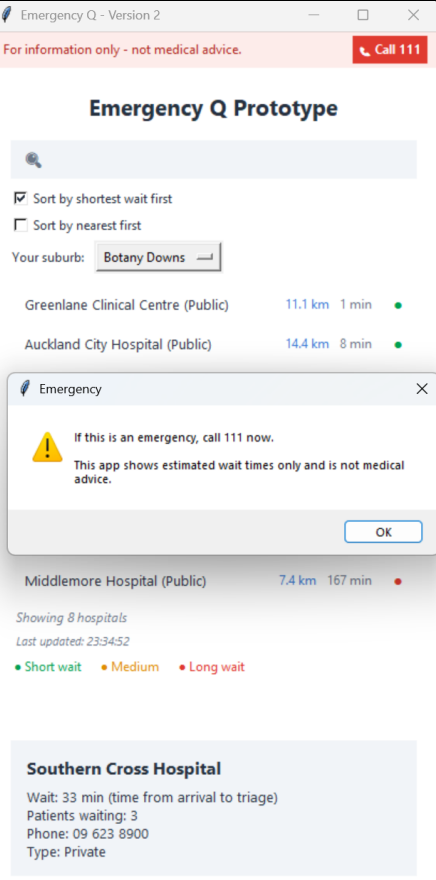
						
Search - no match	Typed "zzz", which matches no hospital.	A clear "No hospitals found" message and a count of 0, with no crash.	Pass	Showed "No hospitals found" and "Showing 0 hospitals" instead of a blank list or an error.	Boundary	None.

						
Sort by shortest wait	Ticked "Sort by shortest wait first".	Hospitals reorder with the shortest wait at the top and the longest at the bottom.	Pass	The list reordered ascending: Greenlane 4 min at the top down to Middlemore 167 min at the bottom.	Expected	None.

						
Location - distances	Chose "Botany Downs" from the suburb dropdown.	A km distance appears on every hospital row straight away.	Pass	Distances appeared immediately on all rows (e.g. Middlemore 7.4 km nearest, Waitakere 25.4 km farthest) - realistic for Botany Downs.	Expected	None.

						
Details panel / selection	Clicked a hospital row (Southern Cross Hospital).	The row highlights and the details panel fills with the wait + definition, patient count, phone and type.	Pass	Southern Cross highlighted blue and the panel showed "Wait: 33 min (time from arrival to triage)", "Patients waiting: 3", the phone number and "Type: Private".	Expected	None.

						
Call 111 button	Clicked the "Call 111" button in the safety banner.	An emergency warning dialog appears.	Pass	The "Emergency" dialog appeared with "If this is an emergency, call 111 now" and the not-medical-advice reminder. OK closed it and returned to the app.	Expected	None.

						
Sort by nearest first	With Botany Downs chosen, unticked "Sort by shortest wait first" and ticked "Sort by nearest first".	Hospitals reorder with the closest one at the top and the farthest at the bottom.	Pass	The list reordered by distance ascending: Middlemore 7.4 km at the top, through to Waitakere 25.4 km at the bottom (7.4, 11.1, 13.0, 13.4, 14.4, 14.5, 20.8, 25.4 km). Matches the automated distance check.	Expected	None.

Emergency Q - Version 2

For information only - not medical advice.

Call 111

Emergency Q Prototype

☐ Sort by shortest wait first

☒ Sort by nearest first

Your suburb:

Botany Downs

Middlemore Hospital (Public)

7.4 km

122 min

Greenlane Clinical Centre (Public)

11.1 km

41 min

Southern Cross Hospital (Private)

13.0 km

29 min

Mercy Hospital Auckland (Private)

13.4 km

12 min

Auckland City Hospital (Public)

14.4 km

35 min

Starship Children's Hospital (Public)

14.5 km

38 min

North Shore Hospital (Public)

20.8 km

126 min

Waitakere Hospital (Public)

25.4 km

34 min

Showing 8 hospitals

Last updated: 23:41:00

Short wait Medium Long wait

Southern Cross Hospital

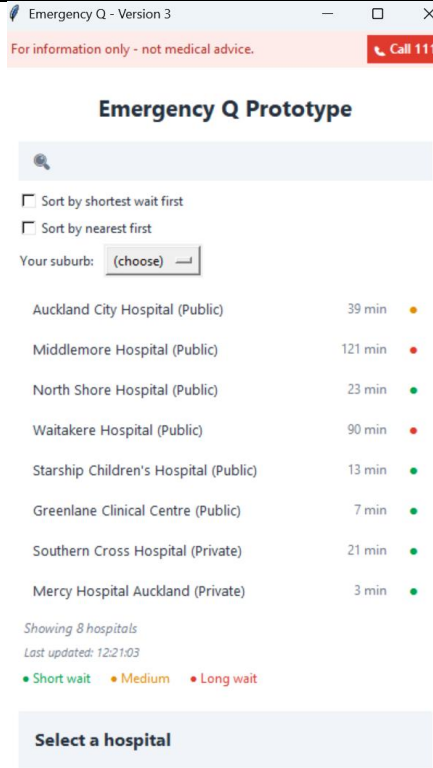
Wait: 33 min (time from arrival to triage)

Patients waiting: 3

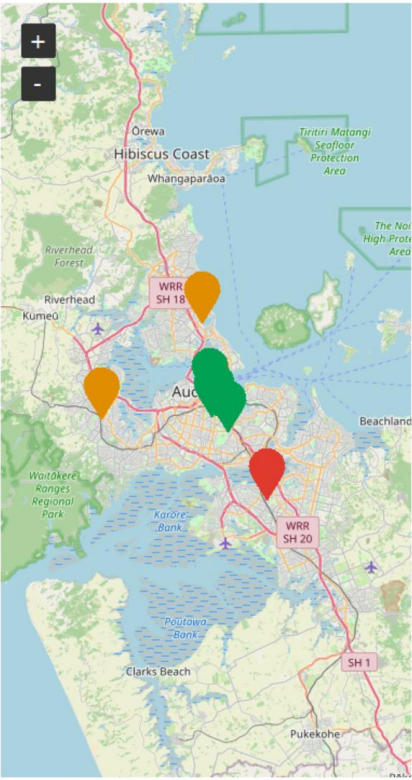
Phone: 09 623 8900

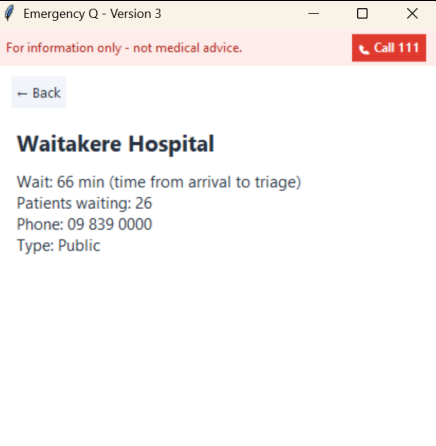
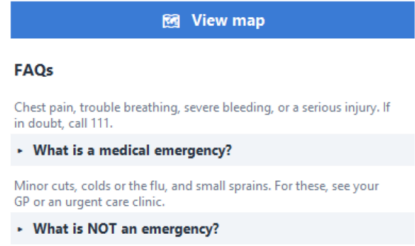
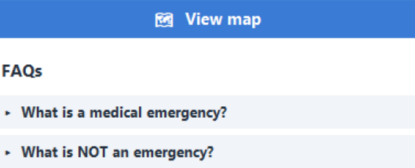
Type: Private

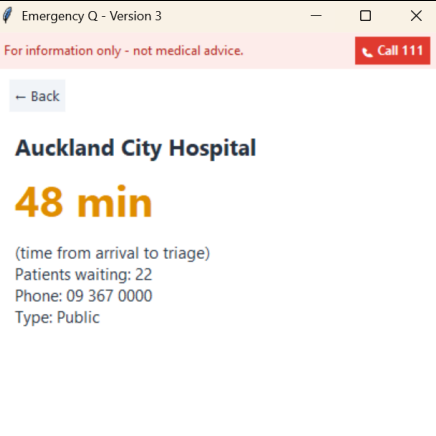
Version 3						
GUI / logic split	Split update_list into get_visible_hospitals (returns the data) + update_list (draws it), then ran the program (Commit 2).	The 8 hospitals load and display as before, now with the logic separated from the drawing.	Fail	Crashed on startup with TypeError: object of type "NoneType" has no len() at str(len(matches)). get_visible_hospitals built the list but had no return, so it handed back None and update_list called len(None).	Expected	Added "return matches" at the end of get_visible_hospitals so it hands the list back to update_list.
GUI / logic split	Re-ran after adding "return matches" (Commit 2).	All 8 hospitals load and display, with the logic now separate from the GUI.	Pass	All 8 hospitals loaded and displayed exactly as before. update_list now just asks get_visible_hospitals for the data and draws it - a clean interface between the GUI and the underlying logic (targets 91906 Excellence).	Expected	None.

						
Sort toggles (mutually exclusive)	Added on_sort_wait / on_sort_nearest so ticking one sort unticks the other, then ticked "Sort by nearest first" (Commit 3).	Ticking "nearest" stays ticked and unticks "shortest wait".	Fail	"Nearest" immediately unticked itself and could never stay on. A copy-paste bug: on_sort_nearest unticked nearest_var (itself) instead of sort_var.	Expected	Changed on_sort_nearest to untick sort_var (the other box). Now only one sort is active at a time.
Sort toggles (mutually exclusive)	Re-ran and ticked each sort box in turn (Commit 3).	Only one box on at a time - ticking one unticks the other.	Pass	Ticking "nearest" unticked "shortest wait" and vice versa. The two sorts can no longer both be on, removing the V2 ambiguity.	Expected	None.
Map with colour-coded pins	Added an interactive tkintermapview map centred on Auckland with a colour-coded pin per hospital, then ran it (Commits 4-5).	A real Auckland map appears with a green/amber/red pin at each hospital, coloured by wait.	Pass	The OpenStreetMap view loaded showing Auckland, with 8 pins coloured by status_colour (green short / amber medium / red long). Removed the marker text labels to declutter the central cluster. tkintermapview is the third-party-library complex technique for 91906.	Expected	None.

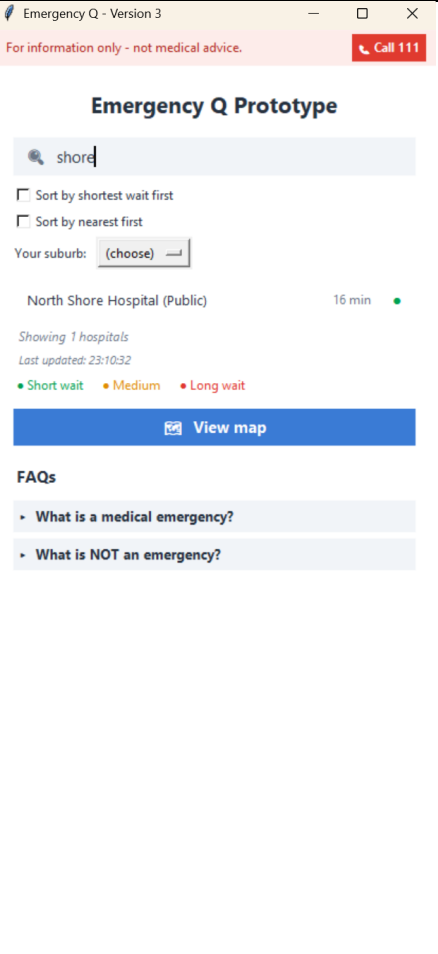
	detail) switched by show_page() with tkraise, then ran it (multi-page Commit 1).	organised into pages.		frame, with a new "View map" button added. 		
Multi-page: map page	Clicked "View map" to open the dedicated map page (multi-page Commit 2).	A full map page opens with the coloured pins and a Back button; Back returns to the search page.	Pass	The map page opened showing the full Auckland map with the 8 colour-coded pins, a title, legend and a Back button, and Back returned to the search page. Added a reusable build_banner() helper so the 111 banner appears on every page with no duplicated code (91906).	Expected	None.

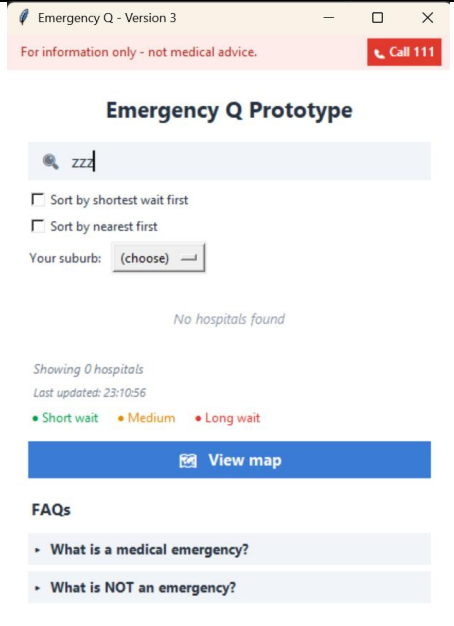
				Emergency Q - Version 3 For information only - not medical advice. Call 111 Back Hospital Map 		
Multi-page: detail page	Clicked a hospital in the list, and a pin on the map (multi-page Commit 3).	Each opens a dedicated detail page for that hospital; Back returns to wherever I came from.	Pass	Clicking a list row or a map pin opened the hospital's own detail page (name, wait, patients, phone, type). Back returned to the search page (from the list) or the map page (from a pin), tracked with a previous_page variable.	Expected	None.

						
Search page FAQs	Added collapsible FAQs to the search page and ran it (multi-page Commit 4).	The FAQ answers start hidden; clicking a question shows/hides its answer.	Fail	The answers showed on startup and clicking a question did nothing. The button was written <code>command=toggle()</code> - the <code>()</code> called <code>toggle</code> immediately (packing the answer open) and handed the button <code>None</code>, so it had no click action. 	Expected	Changed <code>command=toggle()</code> to <code>command=toggle</code> so the function runs on click, not at startup.
Search page FAQs	Re-ran after fixing the command and clicked each question (multi-page Commit 4).	Answers start hidden and toggle open/closed on click.	Pass	The two FAQs now start collapsed and each question expands/collapses its own answer on click. 	Expected	None.
Detail page polish	Made the wait time big and colour-coded on the detail page (multi-page Commit 5).	The wait time is shown large and coloured green/amber/re	Pass	The detail page now shows the wait in large colour-coded text (e.g. Auckland City Hospital "48 min" in amber), matching the list dots and the map pins.	Expected	None.

		d by status_colour.				
Automated logic re- check	Ran the headless logic test suite against the finished V3 file (mocks the GUI + map library and exercises the real code).	All the underlying logic still works after the multi-page refactor.	Pass	27/27 checks passed: loading (8 hospitals/suburbs, patients), colour boundaries (29/30/89/90), distance, search (partial / case-insensitive / no-match / empty), both sorts, mutually-exclusive toggles, nearest-guard, show_details colour-coding, page navigation, and the robust loader (missing file / short line / bad number).	Expected	None.
Search page - full layout	Launched the finished V3 app and checked the search page has every part it should (manual UI test 1).	The search page shows the 111 banner, title, search box, both sort toggles, suburb picker, all 8 hospitals with coloured dots, the count, "Last updated", the legend, a "View map" button and the FAQs.	Pass	Every element was present and correctly laid out. The dot beside each hospital matched its wait (Starship 3 min green, Middlemore 127 min red, Auckland City 62 min amber), matching the legend.	Expected	None.

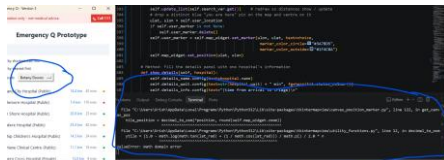
						
Live wait times updating	Watched the search page for ~15 seconds without touching anything (manual UI test 3).	The wait numbers drift up and down, "Last updated" keeps ticking, and no wait goes below 0.	Pass	The wait times nudged a few minutes each refresh and "Last updated" advanced every 3 seconds, simulating a live feed. No value dropped below 0 (guarded by max(0, ...)).	Boundary	None.
Search - partial match	Typed "shore" into the search box (manual UI test 4a).	Only hospitals whose name contains "shore" are listed.	Pass	The list narrowed to just North Shore Hospital and the count read "Showing 1 hospitals". The substring search matched "shore" inside "North Shore" even though it is not at the start of the name.	Expected	None.

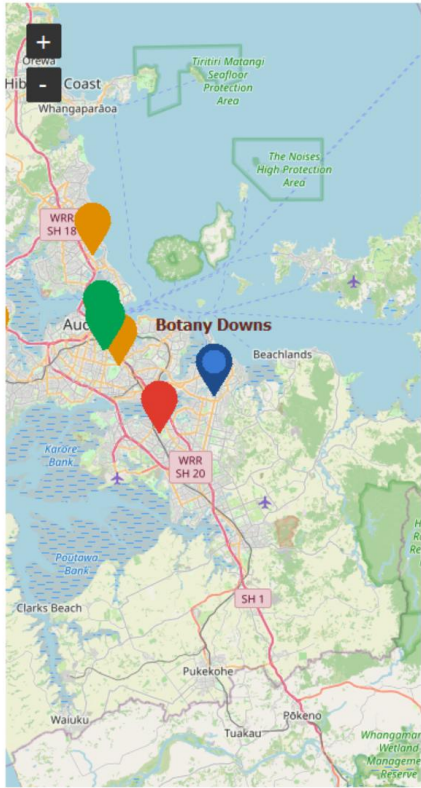
						
Search - no match	Typed "zzz", a string no hospital contains (manual UI test 4b).	The list is empty and a clear "No hospitals found" message shows instead of crashing.	Pass	The list showed "No hospitals found" and the count read "Showing 0 hospitals". Clearing the box brought all 8 back.	Unexpected	None.

						
Sort by shortest wait	Ticked "Sort by shortest wait first" (manual UI test 5).	The list reorders from the shortest wait to the longest.	Pass	The list reordered shortest-first (North Shore 1 min at top through to Middlemore 138 min at the bottom). Ticking this box also unticked "nearest" so only one sort is ever active.	Expected	None.

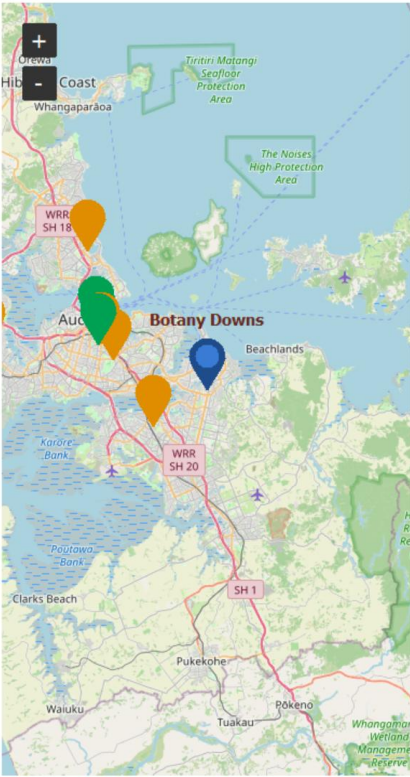
						
Sort by nearest + suburb distances	Chose "Botany Downs" from the suburb picker, then ticked "Sort by nearest first" (manual UI test 6).	A km distance appears on each row and the list sorts from nearest to farthest.	Pass	Each row gained a blue km distance from Botany Downs and the list sorted closest-first (Middlemore 7.4 km at the top through to Waitakere 25.4 km at the bottom). Ticking nearest unticked "shortest wait".	Expected	None.

						
Map pin colour vs detail	While testing the map, clicked North Shore's pin and compared its colour to the wait shown on its detail page (manual UI test).	The pin colour matches the detail page (and the list dot) for the same hospital.	Fail	The pin was amber but the detail page showed the wait as green. place_map_markers() ran only once at startup, so the pins kept their launch colours while update_times kept drifting the waits every 3 seconds - so a pin could sit on a stale colour while the list and detail read the live wait.	Expected	Made place_map_markers() clear the old pins first (stored in self.hospital_markers), and called it from update_times so the pins are recoloured on every live refresh.
Map pin colour vs detail	Re-ran after the fix and watched several pins as the live times drifted across the 30/90 minute boundaries.	Each pin recolours as its wait crosses a boundary and always matches	Pass	The pins now recolour in step with the list every 3 seconds, and a clicked pin's colour matches its detail page. The startup-only staleness is gone.	Expected	None.

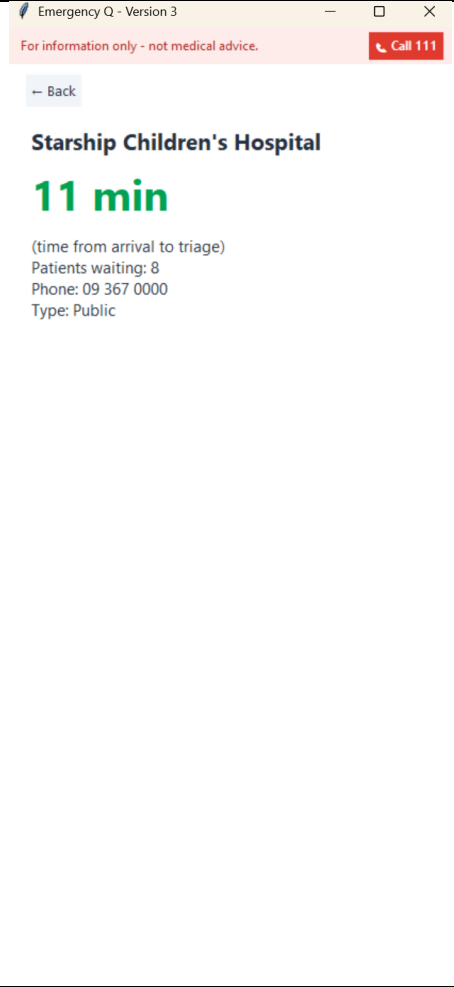
		that hospital's list dot and detail page.				
Show suburb on map	Added a distinct blue "you are here" pin for the chosen suburb, then picked Botany Downs and opened the map (Commit 8).	A blue pin appears on the chosen suburb and the map centres on it.	Fail	Crashed with ValueError: math domain error inside tkintermapview (math.tan on the latitude). set_marker was called as set_marker(ulon, ulat) - the arguments were swapped, so the longitude (~174) was used as a latitude, which is outside the valid -90..90 range and broke the tan calculation. 	Unexpected	Swapped them back to set_marker(ulat, ulon) so it matches the (lat, lon) order set_position already uses.
Show suburb on map	Re-ran after fixing the argument order, picked Botany Downs and opened the map (Commit 8).	A blue pin sits on the chosen suburb and the map centres on it.	Pass	A blue "Botany Downs" pin appeared exactly on the suburb and the map re-centred on it, clearly distinct from the green/amber/red hospital pins. Picking a different suburb moves the single blue pin (old one deleted first).	Expected	None.

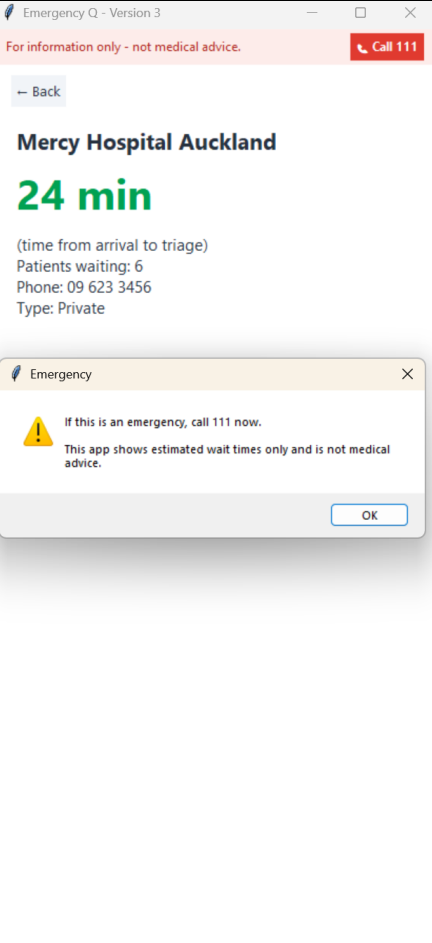
				Emergency Q - Version 3 For information only - not medical advice. Call 111 Back Hospital Map 		
FAQ expand / collapse	Clicked each FAQ question on the search page, then clicked it again (manual UI test 7).	Each answer stays hidden until its question is clicked, then hides again on a second click.	Pass	Clicking "What is a medical emergency?" revealed its answer and clicking again collapsed it; the two FAQs toggle independently and both start hidden on launch.	Expected	None.

						
Navigation - map round-trip	Clicked "View map", checked the map page, then clicked Back (manual UI test 8).	The map page opens with the coloured pins, and Back returns to the search page.	Pass	The map page opened with all the hospital pins plus the blue suburb pin, and "Back" returned to the search page.	Expected	None.

				Emergency Q - Version 3 For information only - not medical advice. Call 111 Back Hospital Map  Short wait Medium Long wait		
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Navigation - list row to detail	Clicked a hospital in the list (Starship), viewed its detail page, then clicked Back (manual UI test 9).	The hospital detail page opens with the big colour-coded wait, and Back returns to the search page.	Pass	Clicking Starship opened its detail page showing "11 min" in green with patients/phone/type, and Back returned to the search page (previous_page = search).	Expected	None.

						
Navigation - map pin to detail	On the map, clicked a pin (Mercy Hospital), viewed its detail page, then clicked Back (manual UI test 10).	A pin opens that hospital detail page, and Back returns to the MAP, not the search page.	Pass	Clicking a pin opened Mercy Hospital's detail page (24 min, green) and Back returned to the map page, tracked by previous_page = map. The 111 button on the detail page also showed the emergency popup.	Expected	None.

						
Call 111 on every page	Pressed the "Call 111" button on the search page, the map page and a detail page (manual UI test 11).	The same emergency warning popup appears on all three pages.	Pass	The emergency popup ("If this is an emergency, call 111 now...") appeared from the shared banner on the search page, the map page and the detail page - confirming the reused build_banner() works on every page.	Expected	None.

				Emergency Q - Version 3 For information only - not medical advice. Call 111 Back Mercy Hospital Auckland 24 min (time from arrival to triage) Patients waiting: 6 Phone: 09 623 3456 Type: Private Emergency ! If this is an emergency, call 111 now. This app shows estimated wait times only and is not medical advice. OK		
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				Emergency Q - Version 3 For information only - not medical advice. Call 111 Emergency Q Prototype ☐ Sort by shortest wait first ☐ Sort by nearest first Your suburb: Botany Downs Auckland City Hospital (Public) 14.4 km12 min Middlemore Hospital (Public) 7.4 km57 min North Shore Hospital (Public) 20.8 km9 min Emergency If this is an emergency, call 111 now. This app shows estimated wait times only and is not medical advice. OK Showing 8 hospitals Last updated: 23:42:15 Short wait Medium Long wait View map FAQs What is a medical emergency? Chest pain, trouble breathing, severe bleeding, or a serious injury. If in doubt, call 111. What is NOT an emergency?		
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